CRH-CFG: Constraint-aware Receding-Horizon Guidance for Attribute-Conditioned Flow Matching

Mu Chen | CMU 10-799 Diffusion & Flow Matching | Controllability Track

One affine guidance axis ✦ one analytic endpoint map turns fixed CFG into constrained feedback.

1-D CFG control authority

x0 analytic endpoint forecast

+0 extra U-Net calls for selection

First principles: derive the controller

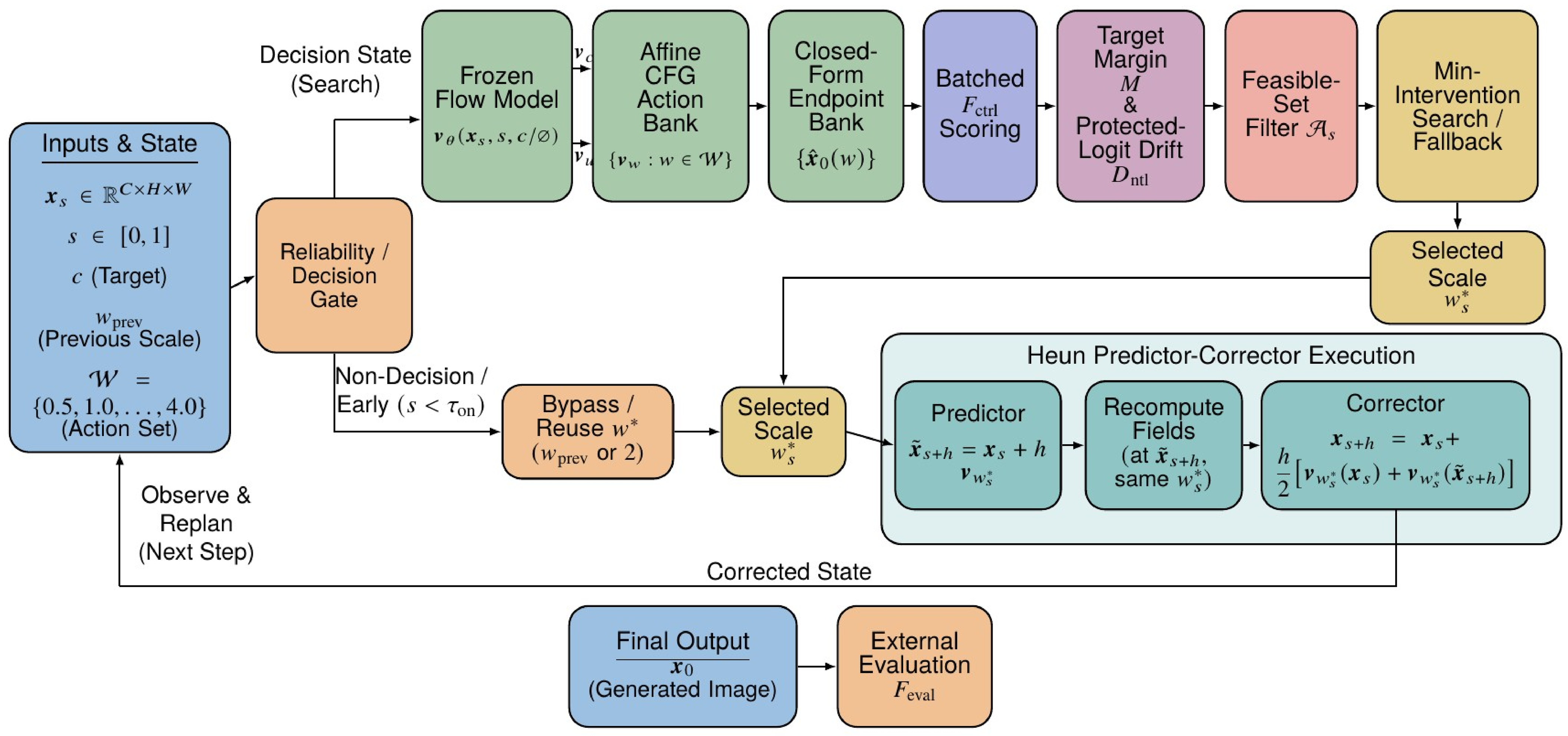
1 Field structure
All scalar CFG actions reuse .

2 Endpoint geometry
A straight path maps each action to a clean-endpoint forecast.

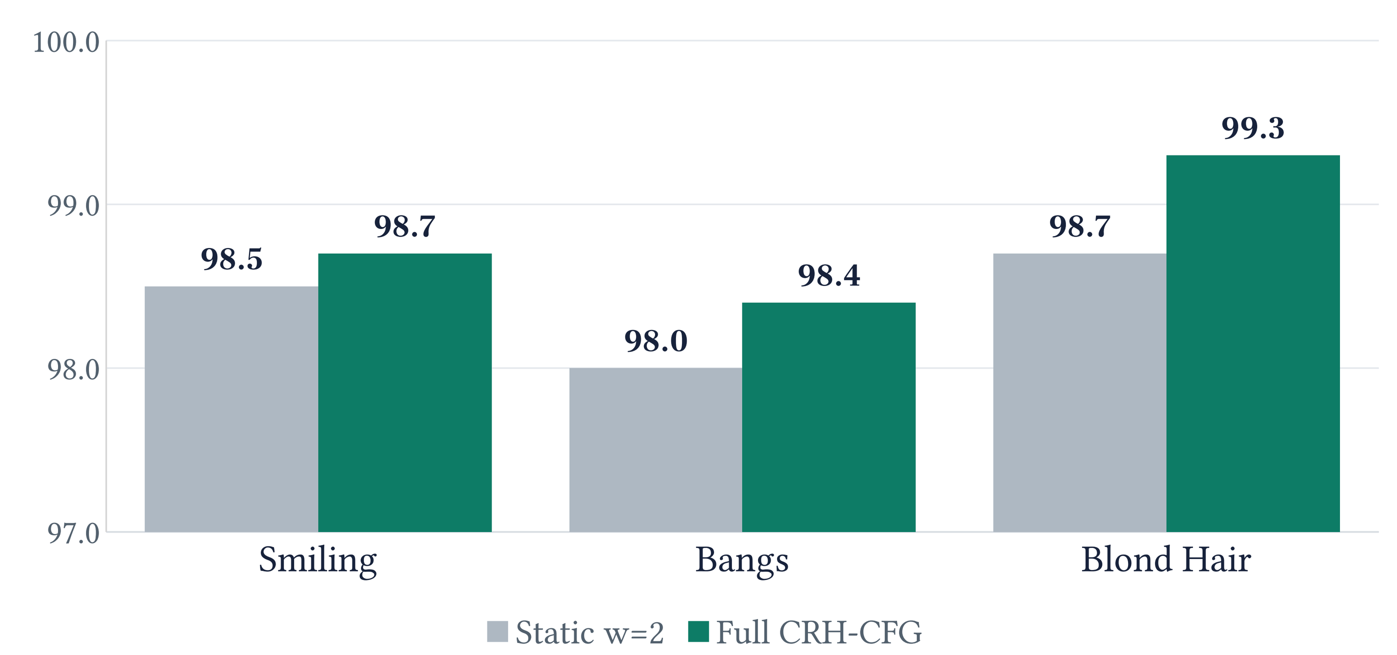
3 Control principle
Satisfy target + protected-attribute proxies with the least intervention, then re-observe.

Fixed w is open-loop. CRH-CFG asks the finite constrained question again at each observed state.

CRH-CFG: field reuse -→ endpoint reasoning -→ constrained action



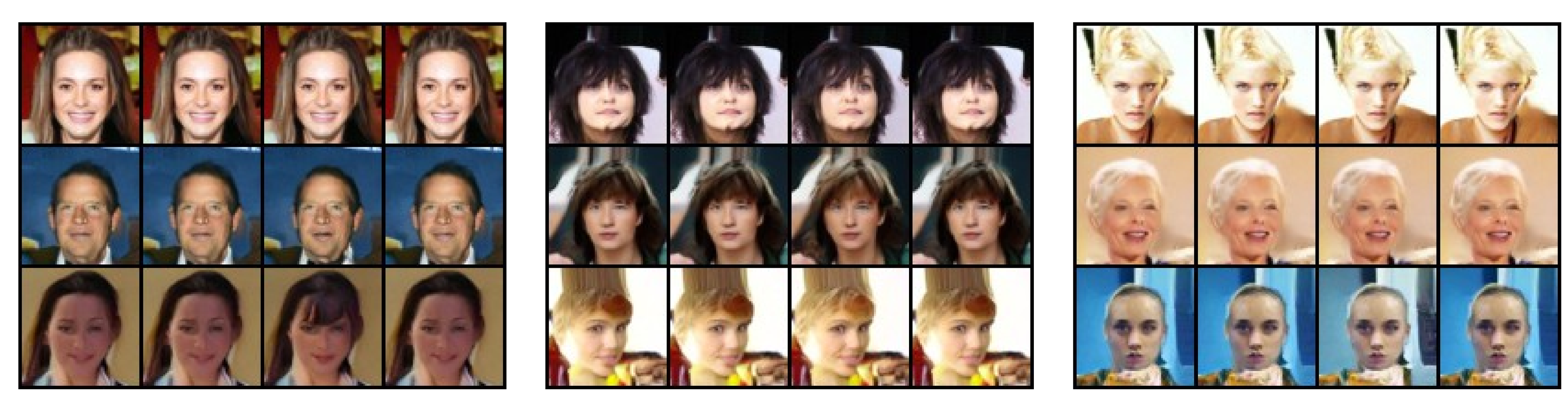
Evidence: adherence at matched generator work



Observed successes (out of 1,000):
Smiling 985 -> 987 | Bangs 980 -> 984 | Blond 987 -> 993

Claim -> evidence -> boundary
Mean success: .9840 -> .9880 (+0.4 pp). Cost: +1.80%, same U-Net work.
Common-w=1 preservation does not improve; KID_marg rises for all targets but cannot identify conditional quality. No per-seed transitions: descriptive only.

Evidence boundary: success, boundary, and failure cases

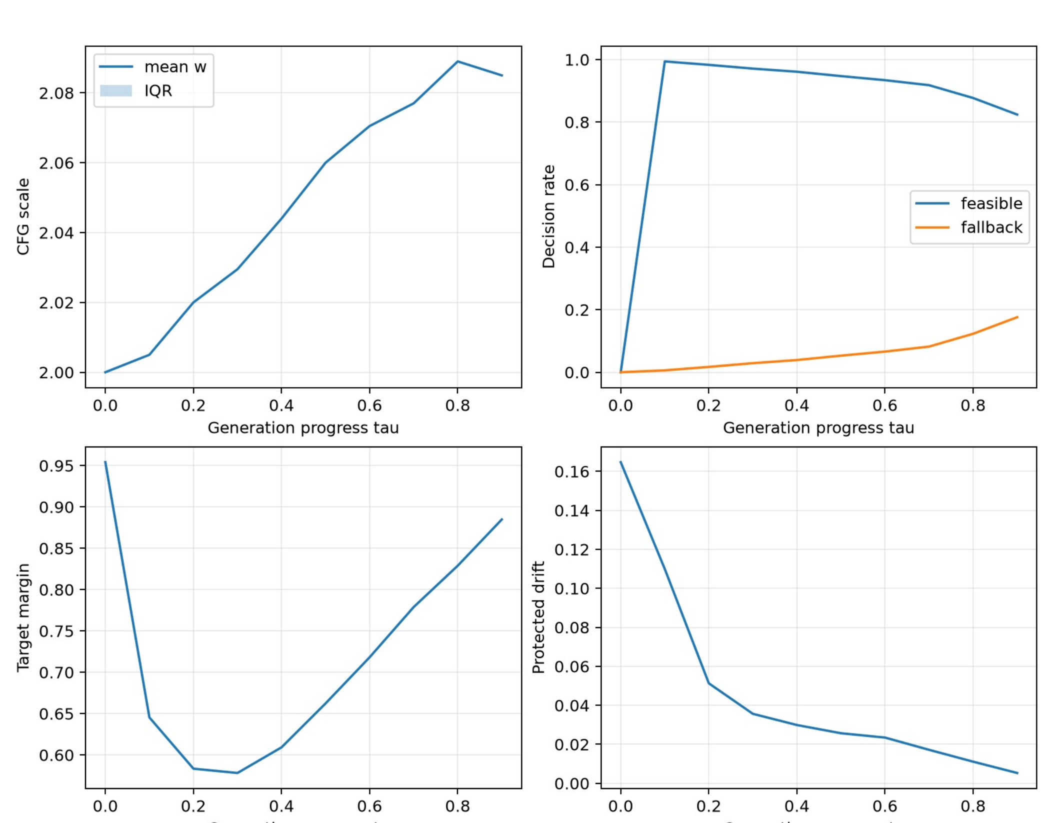


Smiling **Bangs** **Blond Hair**

Columns: static w=2, interval CFG, target-only, full CRH-CFG. Rows are selected by frozen rules: maximum-margin success, nearest evaluator boundary, minimum-margin failure.

Reachability	One scalar direction cannot make every constraint feasible.
Proxy activity	The selected preservation budget never changes the finite-set argmin.
Measurement	Endpoint scores from F_ctrl do not guarantee final outcomes under F_eval.

Adaptation concentrates on difficult trajectories



At the last saved decision (s=0.9):

Smiling mean w 2.017 | fallback 5.7%
Bangs mean w 2.085 | fallback 17.6%
Blond mean w 2.186 | fallback 1.6%

Median and IQR remain w=2 at every saved decision. CRH-CFG mainly corrects a minority of hard samples.

What the ablations and failures teach us

- **Reliability gate is essential**
  Removing it drops reduced-protocol success from 98.83% to 91.54%.
- **Preservation budget is non-binding**
  Adding the preservation term changes no reported reduced-controller value.
- **Extra scale saturates**
  2 -> 3 adds +0.26 pp; 3 -> 4 adds 0.00 pp.
- **Scalar authority is limited**
  Selected failures reach high fallback or w=4 without crossing the evaluator boundary.

Supported: low-overhead adaptive correction. Not supported: joint adherence-preservation-quality improvement.

Reproducibility & compute

3 targets x 1,000 paired seeds | frozen checkpoint/split | 19 tests
HW4: 3.824 measured L40S GPU-h | HW3 training: 2.688 estimated | qualified total: 6.512 GPU-h

Selected references

Lipman et al., Flow Matching, 2023 | Ho & Salimans, CFG, 2022 | Guided Flows, 2023
Limited Guidance Interval / CADS, 2024 | Online feedback, 2025 | CFG++ / Rectified-CFG++, 2024-25

Resources & disclosure

Built independently from the course starter repository and cited papers. OpenAI Codex assisted implementation, analysis, writing, and poster production; all displayed numbers were checked against retained artifacts.